INCH-POUND

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SUPERSEDING
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MILITARY SPECIFICATION

DEHUMIDIFIER, DRY DESICCANT TYPE, AUTOMATIC, CONTINUOUS DUTY

This specification is approved for use by the Naval Facilities Engineering Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for dry desiccant, shallow bed dehumidifier, reactivated automatically.

1.2 Classification. The dehumidifier shall be of the following reactivating heat types, as specified (see 6.2):

Type I - Electric.
Type II - Gas.
Type III - Steam.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer (Code 156), Naval Construction Battalion Center, 621 Pleasant Valley Road, Port Hueneme, CA 93043-4300, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 4440

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SPECIFICATIONS

FEDERAL

PPP-P-40 - Packaging and Packing of Hand Tools.
PPP-T-60 - Tape: Packaging, Waterproof.
PPP-B-636 - Boxes, Shipping, Fiberboard.
TT-P-664 - Primer Coating, Synthetic, Rust-Inhibiting, Lacquer Resisting.

MILITARY

MIL-P-116 - Preservation, Methods of.
MIL-B-121 - Barrier Material, Greaseproofed, Waterproofed, Flexible.
MIL-V-173 - Varnish, Moisture-and-Fungus-Resistant (for Treatment of Communications, Electronic, and Associated Equipment).
MIL-D-3716 - Desiccants, Activated for Dynamic Dehumidification.
MIL-E-16298 - Electric Machines Having Rotating Parts and Associated Repair Parts: Packaging of.
MIL-E-17555 - Electronic and Electrical Equipment, Accessories, and Repair Parts: Packaging and Packing of.
MIL-B-22191 - Barrier Materials, Transparent, Flexible, Heat Sealable.
MIL-P-46093 - Primer Coating, Synthetic (for Brake Drums).

STANDARDS

FEDERAL

FED-STD-595 - Colors.

MILITARY

MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-1186 - Cushioning, Anchoring, Bracing, Blocking and Waterproofing; with Appropriate Test Methods.
MIL-STD-2073-1 - DoD Material Procedures for Development & Application of Packaging Requirements.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents which are current on the date of the solicitation (see 6.2)

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NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

NEMA ICS 1 - General Standards for Industrial Controls and Systems.
NEMA MG 1 - Motors and Generators.

(Application for copies should be addressed to the National Electrical Manufacturers Association, 2101 L Street, N.W., Washington, DC 20037.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 70 - National Electrical Code.

(Application for copies should be addressed to the National Fire Protection Association, 1 Batterymarch Park, P.O. Box 9101, Quincy, MA 02269-9101.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. This specification covers a dry-type desiccant dehumidifier (hereinafter called "dehumidifier") having the desiccant reactivated automatically. The minimum size dehumidifier shall be capable of removing 500 pounds of water per day from air entering the machine at 70 degrees Fahrenheit (F) dry bulb temperature and a relative humidity of 35 percent. The dehumidifier shall consist essentially of the desiccant beds, desiccant, adsorption-air, and reactivation-air fans, fan motors, cycling mechanism, reactivation heaters, controls, protective devices, framework, and casings. Each unit shall be mounted with its auxiliaries on a common base which shall be provided with skids and a slinging eye attached to each corner of the common base for convenient moving. The machines are intended for indoor service and shall be furnished without special casing for waterproofing. The dehumidifying machines shall be electrically controlled. No water cooling service shall be required. Each machine shall be a dynamic type in which humid air is drawn into the machine, passed through the moisture removal device, and the dried air discharged into the space to be dehumidified.

3.2 Standard commercial product. The dehumidifier shall, as a minimum, be in accordance with the requirements of this specification and shall be the manufacturer's standard commercial product. Additional or better features which are not specifically prohibited by this specification but which are a part of the manufacturer's standard commercial product, shall be included in the dehumidifier being furnished. A standard commercial product is a product which has been sold or is being currently offered for sale on the commercial market through advertisements or manufacturer's catalogs, or brochures, and represents the latest production model.

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3.2.1 Measuring and indicating device calibrations. Unless otherwise specified, either the US or SI shall be used to graduate measuring and indicating devices such as scales, depth stops, carriage stops, dial indicators, pressure gauges, temperature indicators, and other similar devices. When only one system of graduation is acceptable, the particular graduation required shall be as specified (see 6.2). Regardless of the measurement system used, all measuring and indicating devices on the machine shall be graduated in the same system.

3.3 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.2.1, 6.2, and 6.4).

3.4 Materials. Materials used shall be free from defects which would adversely affect the performance or maintainability of individual components or of the overall assembly. Materials not specified herein shall be of the same quality used for the intended purpose in commercial practice. Unless otherwise specified herein, all equipment, material, and articles incorporated in the work covered by this specification are to be new and fabricated using materials produced from recovered materials to the maximum extent possible without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. Unless otherwise specified, none of the above shall be interpreted to mean that the use of used or rebuilt products is allowed under this specification.

3.5 Design. The dehumidifier shall be a shallow dry-desiccant type that automatically reactivates the desiccant and provides continuous operation. The unit shall be designed to permit easy accessibility for maintenance and service in use. The design shall be such as to prevent conditions hazardous to personnel or deleterious to equipment.

3.6 Capacity. The dehumidifier shall be capable of removing water at the rates indicated in table I.

TABLE I. Water removal capacity.

Entering air conditions		Relative humidity (percent)	Minimum water removal rate (pounds per day)
Dry bulb Degrees F	Dew point Degrees F		
70	41	35	500
55	28	35	296

3.7 Reactivity. Reactivation cycle time shall not exceed 95 percent of the adsorption cycle when humid air is 70 degrees F or higher.

3.8 Energy consumption. The reactivation energy per pound of water absorbed when supplied with humid inlet air of 35 percent relative humidity shall not exceed the amounts shown in table II.

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TABLE II. Reactivation energy.

Entering air (Dry bulb degrees F)	Electric Kw - Hr	Gas (Btu/h)	Steam-pounds per hour at 100 psi
70	1.00	4000	3.00
40	1.20	7000	4.50

3.9 Desiccants. The desiccant shall be in accordance with type I or II, grade H, of MIL-D-3716, except that the minimum water vapor adsorption capacity percentage shall be as indicated in table III.

TABLE III. Desiccant adsorption capacity.

Normal relative humidity (percent)	Minimum water vapor adsorption (percentage)
10	5.7
20	9.0
40	14.0
60	24.0
80	30.0

3.10 Desiccant bed. The desiccant bed shall be not less than 4 inches in depth and shall be enclosed in an uncoated, corrosion-resistant metal, or cadmium-plated steel screen of sufficient strength to support the bed without deformation. The screen mesh size shall prevent loss of desiccant. The screen shall be supported to prevent collapse or change of shape in normal usage or while being moved. In machines having horizontal air flow through the bed, the desiccant container shall have a reserve of not less than 15 percent additional capacity above the air stream to insure a full bed in the active zone. Means shall be provided for replenishing and completely replacing the desiccant without dismantling the machine. To prevent attrition and loss of desiccant, the air velocity through the mean area of the desiccant bed shall not exceed 60 feet per minute.

3.11 Air intake filters, ductwork, and dampers. The weather air and warehouse air shall be filtered before it enters the desiccant bed. The filters shall be the dry cleanable type of rigid construction having non-ferrous or mineral filter media capable of withstanding air velocities up to 600 feet per minute. The filter shall have a nominal thickness of 2 inches and characteristics of high arrestance efficiency, high dirt holding capacity, and cleanability. The velocity through the gross area of each filter shall be not more than 500 feet per minute. Filters shall be so located as to be readily accessible for cleaning or replacement. The adsorption and reactivation air inlets and outlets shall be provided with aluminum or steel flanges made of material with a minimum thickness of 0.0781 inch (US revised standard gage No. 14), and shall be constructed to minimize the transmission of strains imposed by outside duct connections to the working parts of the machine. Inlet and outlet openings and flanges shall be extended from the machine to facilitate horizontal duct connections. Where possible, elbows shall be assembled with adsorption passages turned toward the opposite side from the reactivation connection. Each connection shall be provided with a 0.750-inch coupling for insertion of supply and inlet temperature sensing elements. All air passages and ducting throughout the machine shall be constructed of metal castings, aluminum, or sheet steel with a minimum thickness of 0.0625 inch (US revised standard gage No. 16).

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Joints in all passages shall be welded or connected by bolted flanges and asbestos type or neoprene type gaskets capable of withstanding temperatures exceeding 350 degrees F. Each air opening of the machine shall be closed and sealed at the factory by blind flanges and shall be protected against injury during shipment. The bottom of the reactivation air outlet shall be not less than 6 inches above the base of the unit to permit grading of the reactivation air line to the outside of the dehumidified space. Insulation shall be provided where necessary to prevent excessive heat losses, and proper means shall be provided to permit drainage of all condensate which may accumulate in areas which do not drain to the air outlets of the machine. Insulation applied to outside surfaces shall be protected with an aluminum or galvanized sheet-metal jacket. Insulation installed within the air passages shall also be protected by an aluminum or galvanized sheet-metal jacket.

3.12 Reactivation heaters. Reactivation shall be accomplished by means of electric, gas, or steam heaters. The heating surface shall be installed in the air passages to the desiccant beds and the maximum temperature shall not exceed 1600 degrees F in normal operation. Automatic control shall be provided to prevent the heater from being energized unless the reactivation fan is delivering air to the desiccant bed. In order to protect the heater bank from overheating in the event of insufficient flow of reactivation air, an over-limit switch shall be provided to shut off the heater. The switch shall reset automatically when the flow of reactivation air has returned to normal. Design of the heating element shall be such that there will be no points where high concentration of heat can occur.

3.12.1 Electric. Electric heaters for type I dehumidifiers may be of the encased wire, strip, finned metal-sheathed, open coil, or open ribbon type, and shall be suitable for forced air heating. Unless otherwise specified (see 6.2), heaters shall operate on a 208 volt, 3-phase, 60 Hertz (Hz) system. All electrical junctions to or in the heater element shall be secured in such a manner that the resistance of the junctions will not cause the junction to heat to a temperature exceeding the temperature of the heating element under normal operation. The heater shall be capable of withstanding indefinitely a temperature of 25 percent in excess of normal operating temperatures. The electric energy demand of each machine shall not exceed 25 kilowatts, and shall be divided between heaters and fan motors. Heaters shall be installed in such a manner that inspection, servicing, and replacement is facilitated without dismantling the machine.

3.13 Valves and valve-actuating motors. Valves for the control of air flow (humid and reactivation) through the desiccant beds shall be of the type best suited for the service. Valves shall be constructed of materials which will give maximum resistance to corrosive action of moist air and a minimum of wear of operating parts, and shall be ribbed or otherwise reinforced to prevent distortion resulting from temperature changes. Automatic or manual adjustment shall be provided for takeup for wear. The valve construction shall be of sufficient rigidity and strength to prevent deformation under shipment, installation, or normal operation. Valve operators shall be valve-actuating motors of a type and capacity required for changing the valve position to reverse the air flow. Provision shall be made to prevent damage to the valve-actuating motor in the event of the sticking of the valve. The valve movement operation shall be completed without undue lapse of time. The construction of the machine shall be such that bypassing of humid air from the

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reactivation air passages to the adsorption air passages will be negligible during the period of valve switching. Valve plug or disk and body seat faces shall be a smooth finish and shall be provided with means for positive lubrication to eliminate binding or sticking. If fluid lubricant for valves is required, a supply reservoir showing amount of lubricant available, or other means of showing sufficient lubricant shall be provided. Valves shall be so arranged on the machine as to be removable for repair or replacement without movement of the machine or dismantling any of the major parts of the unit.

3.14 Fans or blowers. Fans or blowers shall be of the single or multi-stage centrifugal type or of the axial-flow type, and shall be capable of circulating humid air through the machine from the space to be dehumidified and reactivation air obtained from the outside atmosphere. Fans or blowers shall have capacities sufficient to meet the requirements of this specification. Fan wheels shall be dynamically balanced at operating speed. Clearances between moving and stationary parts shall be ample for the service. Fan or motor bearings shall be the ball type and shall be easily accessible for lubrication.

3.15 Fan motors and motor starters. Motors for fans and blowers shall be the conventional squirrel-cage, open, continuous-duty induction type. Unless otherwise specified (see 6.2), the motors shall be designed for 200 volt, 3-phase, 60 Hz service. Motors shall conform to NEMA MG 1 and motor starting equipment shall conform to NEMA ICS 1. All motors shall be standard NEMA frame motors and the fans and motors shall be statically and dynamically balanced. Three-phase motor branch circuit protection shall be provided for each motor and motor branch circuit conductors shall be capable of carrying 125 percent of the full-load motor current. Motor starters shall be of the magnetic full-voltage across-the-line type and shall be provided with low-voltage release and overload protection. The starter power contacts shall be capable of carrying the full-load current of the motor. The starting equipment for the reactivation fan shall be interconnected with the no-air-flow mechanism and the reactivation cycle control. The horsepower rating of each motor shall be ample for the service specified and the motor speed selected shall be best suited for the operation. Motors shall be belt-driven or direct-connected to the fans. All power lines interconnecting the motors and power panel mounted on the unit shall have flexible connections to eliminate the transmission of vibrations.

3.16 Adsorption cycle control. When specified (see 6.2), each machine shall dehumidify the air during the adsorption cycle either continuously or intermittently, as the case may be, in response to the functioning of an external control. The length of the adsorption cycle in each machine shall be adjusted manually through an adsorption timer which shall be provided with an adjusting dial having a range from zero to not less than 12 hours and graduated in quarter hours. At the end of the adsorption cycle as indicated by the dial on the timer, the power controllers shall be actuated so as to cause all valves in the air passages to be reversed, the reactivation air to be passed through the proper desiccant bed, and the reactivation heater to be energized, except in the case where at the end of the adsorption cycle, the other desiccant bed of the machine is still reactivating. In this case, the bed shall stay on adsorption service until reactivation has been completed on the other bed.

3.17 Reactivation cycle control. When specified (see 6.2), the reactivation cycle shall be controlled in each machine by a cycle timer to energize the heaters when the adsorption cycle has been terminated. The length of the

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reactivation cycle shall be adjusted manually through a cycle timer which shall be provided with an adjusting dial having a range of zero to not less than 120 minutes and graduated in minutes. Before the end of the reactivation cycle as indicated by the dial on the timer, the reactivation heaters shall be de-energized, and the flow of reactivation air shall continue for a predetermined period to purge the bed with reactivation air in order to remove the moisture released by the desiccant as the bed is being cooled. At the end of the purging period, the fan shall be de-energized, and the necessary electrical connections shall be made to permit the return to adsorption service. After reactivation has started, it shall be continued to completion even though the external control shuts down the adsorption cycle on the other desiccant bed.

3.18 Power control panel. All control equipment, except the line switch, shall be mounted in a power control panel fastened to each unit. The panel shall be not less than 0.1875-inch sheet steel and shall have a sheet steel enclosure fitted with a door equipped with hinges and a built-in cylinder lock. A pocket shall be provided to hold a copy of an instruction booklet. The panel shall be equipped with the following:

a. Manual momentary start and stop switch to actuate circuit breaker.
b. A 208/110 volt transformer to provide 110 volts for control circuit.
c. Single pole tumbler switch to short-circuit humidity control leads for manual operations.
d. Manually adjustable adsorption and reactivation timers.
e. Running-time meter.
f. Cycle counters.
g. Necessary relays, contactors, and fuse protectors for control circuit.

The control circuit shall be so arranged that any one overload protection device shall interrupt the entire operation. All electrical equipment shall be accessible for servicing. All wiring for the equipment shall be in accordance with NFPA 70. Provisions shall be made for adequate grounding of the equipment. Wiring shall be complete and ready for external connections.

3.19 Circuit breaker. Each machine shall be provided with a 3-pole circuit breaker of the quick-make quick-break type, for disconnecting the unit from the power supply circuit and for motor branch circuit protection. The circuit breaker shall be detached from the power control panel and shall be mounted in its own enclosure adjacent to the power panel.

3.20 Cycle counter. Cycle counter shall be operated electrically or mechanically, shall have a dial of four or five figures, and shall register the number of machine cycles which occur during operation.

3.21 Running-time meters. Running-time meters (integrating timers) shall be provided to register the total operating time of the adsorption and reactivation cycles. Each timer shall consist of a 110 volt synchronous motor, which shall be connected through suitable gearing to counters arranged to register, respectively, the total hours of adsorption and reactivation. The time motors shall be installed in parallel with the 110 volt control circuits closing the contactors on adsorption and reactivation, respectively. Meters shall have provision for locking the adjustment.

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3.22 Lifting attachments. When specified (see 6.2), suitable lifting attachment plate(s) shall be provided to enable the equipment to be lifted in its normal position. Attachment eye(s) shall be not less than 3 inches in diameter, and plate(s) shall withstand any handling conditions encountered, such as rapid lowering and braking of the load. When practicable, only one attachment shall be used. Where more than one is required, each attachment, when practicable, shall be of sufficient capacity to carry the total weight. A copper- or zinc-base alloy, or aluminum plate, securely fastened to the unit, shall indicate diagrammatically the lifting capacity of each attachment.

3.23 Lubrication. All parts that require lubricants shall be provided with suitable means for lubrication.

3.24 Fungus resistance. When specified (see 6.2), electrical components and circuit elements, including terminal and circuit connections, shall be coated with varnish conforming to MIL-V-173, except that:

a. Components and elements inherently inert to fungi or in hermetically sealed enclosures need not be coated.
b. Current-carrying contact surfaces, such as relay contact points, shall not be coated.

3.25 Treatment and painting. The unit (except for metal parts subject to high temperatures) shall be painted in accordance with good commercial practice. The color shall be No. 16187, gray gloss, conforming to FED-STD-595.

3.26 Dissimilar metals. Metals dissimilar with respect to the galvanic scale shall not be used unless separated by an insulating material which will avoid electrolytic corrosion. Table IV groups metals that are compatible.

TABLE IV. Grouping of metals.

Group I	Group II	Group III	Group IV
Magnesium alloys	Aluminum	Zinc	Copper and its alloys
	Aluminum alloys	Cadmium	Nickel and its alloys
	Cadmium	Steel	Chromium
Most anodic	Tin	Lead	Stainless steel
	Stainless steels	Tin	Gold
		Stainless steels	Silver
			Most cathodic

3.27 Identification marking. Identification shall be permanently and legibly marked directly on the dehumidifier or on a corrosion-resisting metal plate securely attached to the dehumidifier at the source of manufacture. Identification shall include the manufacturer's model and serial number, name and trademark to be readily identifiable to the manufacturer.

3.27.1 Equipment and assembly labels, identification and instruction plates. Equipment and assembly labels, identification and operating instruction plates shall be permanently and legibly marked directly on corrosion-resisting plate(s) and securely attached to the dehumidifier. The following individual identification and instruction plates shall be provided:

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a. "Dry air outlet."
b. "Humid air inlet." (Returned air from conditioned space)
c. "Reactivation air inlet." (Outside air in-take)
d. "Reactivation air outlet." (Wet air exhaust)
e. "Filter inspection every 2 months."
f. "Warning." (In red or vivid orange) -- "Check for proper phase before starting machine. Check rotation of motors after starting machine."
g. "Connect power leads here."
h. "Connect humidity-control leads here."
i. "Feeder switch."
j. "Control box."
k. "Manual-control circuit switch."
l. "Warning." (In red or vivid orange) -- "Must be open for automatic operation."

The power panel shall include the name of each item or symbol adjacent to the terminal so that it can be located readily from the wiring diagram.

3.28 Spare parts and maintenance tools. When specified (see 6.2), spare parts and maintenance tools shall be furnished.

3.29 Workmanship.

3.29.1 Metal fabrication. Metal used in fabrication of equipment shall be free from kinks and sharp bends. The straightening of material shall be done by methods that will not cause injury to the metal. Shearing and chipping shall be done neatly and accurately. All bends of a major character shall be made with controlled means in order to insure uniformity of size and shape.

3.29.2 Bolted connections. Bolt holes shall be accurately punched or drilled and shall have the burrs removed. Washers or lockwashers shall be provided in accordance with good commercial practice, and all bolts, nuts, and screws shall be tight.

3.29.3 Riveted connections. Rivet holes shall be accurately punched or drilled and shall have the burrs removed. Rivets shall be driven with pressure tools and shall completely fill the holes. Rivet heads, when not countersunk or flattened, shall be of approved shape and of uniform size for the same diameter of rivet. Rivet heads shall be full, neatly made, concentric with the rivet holes, and in full contact with the surface of the member.

3.29.4 Welding. Welding procedures shall be in accordance with a nationally recognized welding code. The surface of parts to be welded shall be free from rust, scale, paint, grease, or other foreign matter. Welds shall be of sufficient size and shape to develop the full strength of the parts connected by the welds. Welds shall transmit stress without permanent deformation or failure when the parts connected by the weld are subjected to proof and service loadings.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein.

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Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this document where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this document shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in this document shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2.1).
b. Quality conformance inspection (see 4.2.2).

4.2.1 First article inspection. First article inspection shall be performed on one dehumidifier when a first article is required (see 3.3 and 6.2). This inspection shall include the examination of 4.3 and the tests of 4.4. The first article may be either a first production item or a standard production item from the supplier's current inventory provided the dehumidifier meets the requirements of the specification and is representative of the design, construction, and manufacturing technique applicable to the remaining dehumidifiers to be furnished under the contract.

4.2.2 Quality conformance inspection. Quality conformance inspection shall include the examination of 4.3, the tests of 4.4.2 and 4.4.3, and the packaging inspection of 4.5.

4.3 Examination. Each dehumidifier shall be examined for compliance with the requirements specified in section 3 of this document. Any redesign or modification of the contractor's standard product to comply with specified requirements, or any necessary redesign or modification following failure to meet specified requirements shall receive particular attention for adequacy and suitability. This element of inspection shall encompass all visual examinations and dimensional measurements. Noncompliance with any specified requirement shall constitute one defect.

4.4 Tests. The dehumidifier shall be tested, and any unit failing to pass the following tests, as applicable, shall be rejected. Tests shall be conducted as outlined in the referenced documents and as specified herein.

4.4.1 Electrical. Each electrical circuit shall be tested for proper installation and operation of all components. Nonconformance to the requirements of section 3 and NFPA 70 shall constitute failure of the test.

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4.4.2 Operation. The dehumidifier shall be operated continuously for a period of 40 hours under varying air entering conditions specified in 3.6, with all components in service, except that it shall be with or without the heaters. If heaters are not installed during the 40 hour test, immediately following this test, the heaters shall be connected and the machine operated for two complete cycles on each desiccant bed. During the test, there shall be no undue heating of bearings, and the satisfactory operation of control equipment, thermo-switches, thermostats, relays, safety devices, and valves shall be demonstrated. Excessive heating of bearings and unsatisfactory operation of control equipment, thermo-switches, thermostats, relays, safety devices, and valves shall constitute failure of the test.

4.4.3 Water removal capacity. At the end of each 8-hour period of the operation test of 4.4.2, water removed from the air by the dehumidifier unit shall be collected and taken from the unit and weighed to determine the amount of water that has been removed during the past 8-hour period. Nonconformance to 3.6 shall constitute failure of the test.

4.4.4 Air leakage. Air leakage test shall follow immediately the operation test and before the cooling of the unit has taken place. A measured amount of air under a static pressure of not less than 6 inches of water shall be introduced through a gas flow meter to the reactivation or adsorption inlet when the valves are in one of the two normal operating positions and the reactivation outlet blanked off. In lieu of the foregoing, the Government may require that the air supply be introduced through the adsorption inlet, with the adsorption outlet blanked off. The rate of leakage under the tests shall not exceed 3 percent of the rated cubic feet per minute of the machine. Nonconformance to the requirements of 3.11 shall constitute failure of this test.

4.5 Packaging inspection. The preservation, packing, and marking of the dehumidifiers shall be inspected to verify conformance to the requirements of section 5.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or commercial as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Methods of preservation. Cleaning processes, drying procedures, preservatives, and methods of preservation are listed in MIL-P-116, and shall conform to the requirements of MIL-P-116 and any applicable specifications.

5.1.1.2 Disassembly. Disassembly shall be the minimum necessary to protect parts subject to damage or loss, and to accomplish reduction in cube. Removed bolts, nuts, pins, screws and washers shall be reinstalled in mating parts and secured to prevent their loss.

5.1.1.3 Matchmarking. Parts removed and mating parts on the equipment and attachments shall be matchmarked to facilitate reassembly. Parts and accessories removed, and mating parts on the equipment, shall be identified with weatherproof tags attached to mating parts and locations. Markings shall be applied to the tags with a waterproof material.

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5.1.1.4 Cleaning and drying. Prior to the application of preservative compounds or paint, surfaces shall be cleaned by process C-1 and dried by any applicable procedure of MIL-P-116.

5.1.1.5 Unprotected surfaces. Unprotected exterior metal surfaces requiring the application of a contact preservative in accordance with MIL-P-116 and not specifically provided for herein shall be preserved as follows:

5.1.1.5.1 Unfinished (not machined) surfaces. Unfinished exterior metal surfaces shall be coated with type P-1 preservative.

5.1.1.5.2 Machined surfaces. Exposed machined surfaces shall be coated with type P-6 or P-11 preservative and wrapped or covered, as applicable, with barrier material conforming to MIL-B-121, type I, grade A, class 2 or MIL-B-22191, type II. The material shall be secured in place with waterproof tape.

5.1.1.6 Electrical motors. Preserve level A by the alternate method of MIL-E-16298.

5.1.1.7 Electrical equipment. Electrical equipment shall be preserved and packaged in accordance with the level A requirements of MIL-E-17555.

5.1.1.8 Gas heaters and steam heaters. All openings to the interior of the unit shall be covered with barrier material conforming to MIL-B-121, type I or II, grade A, and secured in place with tape conforming to PPP-T-60, type IV.

5.1.1.9 Exposed drive chains. Exposed drive chains shall be coated with enough type P-3, P-9, or P-21 preservative to insure penetration of the preservative to the inner surface of the rollers, pins, and bushings. After the excess preservative has drained, the entire chain and the unpainted surfaces of the sprocket shall be coated with type P-1 preservative.

5.1.1.10 Enclosed chains. Enclosed chains and chain housings shall be preserved by filling the housing to the operating level with the approved lubricant required for operation.

5.1.1.11 Drive belts and pulleys. Drive belts shall be removed or released from tension. Removed belts shall be preserved method IC-1 or IC-3. Unpainted surfaces shall be coated with primer conforming to TT-P-664 or MIL-P-46093.

5.1.1.12 Desiccant. Desiccant for each individual unit shall be packaged in accordance with MIL-D-3716. The desiccant shall be packaged in 5-, 25-, 100- or 200-pound capacity containers, as specified (see 6.2).

5.1.1.13 Lubrication. All parts requiring lubrication and not specifically provided for herein shall be lubricated in accordance with the contractor's standard practice.

5.1.1.14 Tools and repair parts. Tools and repair parts shall be preserved and packaged in accordance with the level A requirements of PPP-P-40, or when items are not specifically covered by PPP-P-40, an applicable submethod of preservation of MIL-P-116 shall be used.

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5.1.1.15 Technical publications. Technical publications for each piece of equipment shall be preserved method IC-1 or IC-3.

5.1.1.16 Consolidation. Loose components, tools, repair parts, and publications for each dehumidifier shall be consolidated in boxes conforming to PPP-B-636, class weather-resistant.

5.1.2 Commercial. The equipment shall be preserved in accordance with the contractor's standard practice in a manner to prevent deterioration and damage. The equipment shall be lubricated for operational service as required in the operator's manual.

5.2 Packing. Packing shall be level A, B, or commercial as specified (see 6.2).

5.2.1 Levels A and B. The complete unit shall be packed in a box conforming with MIL-STD-2073-1, appendix C, table VII, except open crates shall not be used for Level A packing. The gross weight and level of protection shall determine the container selected. Contents shall be cushioned, blocked, braced, and anchored in accordance with MIL-STD-1186. When specified (see 6.2), desiccant shall be packed separately from the dehumidifier specified in MIL-D-3716 for the appropriate requirement, to include palletization.

5.2.2 Commercial. The equipment shall be prepared for shipment in a manner which will insure arrival at destination in a satisfactory condition. Preparation for delivery shall comply with applicable carrier rules and regulations.

5.3 Marking. Marking shall be in accordance with MIL-STD-129.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Dehumidifiers covered in this specification are intended for dehumidification of warehouse or other storage spaces.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Type required (see 1.2).
c. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
d. When either the US or SI system of measurement is required (see 3.2.1).
e. When first article is required for inspection and approval (see 3.3).
f. Power required, if other than as specified (see 3.12.1 and 3.15).
g. When adsorption cycle control is required (see 3.16).
h. When reactivation cycle control is required (see 3.17).
i. When lifting attachment plates are required (see 3.22).
j. When electrical components and circuit elements shall be coated with varnish conforming to MIL-V-173 (see 3.24).
k. When spare parts and maintenance tools should be furnished (see 3.28).

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l. Level of preservation and level of packing required (see 5.1 and 5.2).
m. Size of container in which desiccant is to be packaged (see 5.1.1.12).
n. When desiccant shall be packed separately from dehumidifier, in accordance with MIL-D-3716 (see 5.2.1).

6.3 Data requirements. When this specification is used in an acquisition and data are required to be delivered, the data requirements shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List (DD Form 1423), incorporated into the contract. When the provisions of DoD Federal Acquisition Regulations (FAR) Supplement, Part 27, Sub-Part 27.475-1 are invoked and the DD Form 1423 is not used, the data should be delivered by the contractor in accordance with the contract or purchase order requirements.

6.3.1 Technical manuals. The requirement for technical manuals should be considered when this specification is applied on a contract. If technical manuals are required, military specifications and standards that have been cleared and listed in DOD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL) must be listed on a separate Contract Data Requirements List (DD Form 1423), which must be acquired under separate contract line item in the contract.

6.4 First article. When a first article inspection is required, the item will be tested and should be a first production item or it may be a standard production item from the contractor's current inventory as specified in 4.2.1. The first article should consist of one unit. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, test, and approval of the first article.

6.5 Subject term (key word) listing.

Adsorption
Reactivity
Water removal

6.6 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Navy - YD

User Activity:
Navy - CG

Preparing Activity:
Navy - YD

(Project 4440-0061)

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description, assembly, installation, operation, trouble shooting ... to the

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-D-16886H	930319

3. DOCUMENT TITLE: DEHUMIDIFIER, DRY DESICCANT TYPE, AUTOMATIC, CONTINUOUS DUTY

4. NATURE OF CHANGE (*Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.*)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial	(2) AUTOVON
MR. DANNY C. MUI	805-982-5666	551-5666
c. ADDRESS (Include Zip Code) COMMANDING OFFICER NSWC CODE 1562G, 621 PLEASANT VALLEY ROAD PORT HUENEME, CA 93043-4300	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

DD Form 1426, OCT 89 — Previous editions are obsolete. 198-290

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